

(2 Marks)

1. Explain Goldratt's critical chain.

- **Critical Chain Method (CCM)** is a project management technique developed by **Eliyahu M. Goldratt** based on the **Theory of Constraints**.
- It focuses on identifying the **longest sequence of dependent tasks** in a project while considering **resource availability**.
- The **critical chain** determines the **minimum time required to complete the project**.
- Unlike the **Critical Path Method**, it considers both **task dependencies and resource constraints**.
- It uses **buffers (project buffer, feeding buffer, and resource buffer)** to protect the project schedule from delays.
- It reduces **multitasking** so that resources can focus on completing one task at a time.
- The main goal is to **improve project efficiency and ensure timely project completion**.

2. Write about Risk Response Strategies for Negative Risks (Threats).

Risk response strategies for negative risks are techniques used in project management to **reduce or control the impact of potential threats** that may affect project objectives such as scope, cost, schedule, or quality. These strategies help project managers prepare actions to handle risks effectively.

- **Avoid** : This strategy eliminates the risk completely by changing the project plan, scope, or schedule so that the threat does not affect the project.
- **Mitigate** : It involves taking preventive actions to **reduce the probability or impact** of the risk. For example, using better technology or adding extra testing.
- **Transfer** : The responsibility of the risk is shifted to a **third party**, such as through insurance, outsourcing, or contracts.
- **Accept** : The project team decides to accept the risk and deal with it if it occurs. This may include preparing contingency reserves.
- **Escalate** : If the risk is beyond the authority of the project manager or project team, it is **escalated to higher management** for decision-making.
- **Contingency Planning** : A backup plan is prepared in advance so that if the risk occurs, the project team can respond quickly and minimize its impact.

3. Describe Qualitative and Quantitative Risk Assessment Techniques.

Qualitative Risk Assessment

- **Definition**: Evaluates risks based on their **probability and impact** without using numerical values.
- **Purpose**: Helps **prioritize risks** so the project team can focus on the most important ones.
- **Techniques**: Probability–Impact Matrix, Expert Judgment, Risk Categorization.
- **Nature**: Uses **descriptive scales** such as high, medium, or low.
- **Usage**: Usually performed **early in the project** because it is simple and quick.

Quantitative Risk Assessment

- **Definition**: Analyzes risks using **numerical data and statistical methods**.
- **Purpose**: Measures the **overall impact of risks** on project objectives like cost and schedule.
- **Techniques**: Monte Carlo Simulation, Decision Tree Analysis, Expected Monetary Value (EMV).
- **Nature**: Uses **mathematical models and numerical values**.
- **Usage**: Applied to **high-priority risks** for more detailed and accurate analysis.

4. What is Project Stakeholder Engagement?

- Project stakeholder engagement is the **process of communicating and working with stakeholders** to meet their needs, expectations, and concerns during the project.
- It involves **identifying stakeholders** who can affect or are affected by the project, such as customers, sponsors, and project team members.
- The project team must **understand stakeholders' interests, influence, and expectations** to manage them effectively.
- Proper stakeholder engagement helps in **maintaining regular communication and involvement** in project activities and decisions.
- It helps to **build trust and strong relationships** between the project team and stakeholders.
- Effective stakeholder engagement **improves project success and reduces conflicts or misunderstandings**.

5. Define Project Audit.

- A **project audit** is a systematic and independent examination of a project to evaluate its **performance, progress, and compliance with project plans and standards**.
- It helps to **identify problems, risks, and areas for improvement** in the project.
- Project audit ensures that the project is **being managed effectively and resources are used properly**.
- It reviews different aspects of the project such as **scope, schedule, cost, quality, and management processes**.
- The main objective of a project audit is to **improve project performance and increase the chances of project success**.
- It also helps in **ensuring that project objectives are achieved within the planned time and budget**.

6. What is Resource Levelling? Why is it important?

Resource levelling is a project management technique used to balance the demand for resources with the available supply. It involves adjusting the project schedule, such as delaying or rescheduling tasks, so that resources like manpower, equipment, and materials are not overallocated. The main goal of resource levelling is to ensure that resources are used efficiently throughout the project.

Importance of Resource Levelling

- It helps to **avoid overallocation of resources** in the project.
- It ensures **efficient utilization of available resources**.
- It helps in **reducing conflicts when the same resource is required for multiple tasks**.
- It improves **project scheduling and workflow management**.
- It helps in **completing the project more efficiently and with better planning**.

(5 Marks)

7. Explain Earned Value Management (EVM) techniques for measuring project performance.

Earned Value Management (EVM) is a project management technique used to **measure project performance by comparing planned work, actual work completed, and actual cost incurred**. It integrates **scope, schedule, and cost** to evaluate whether the project is **ahead or behind schedule and under or over budget**. It also helps project managers take corrective actions when necessary.

EVM Techniques for Measuring Project Performance

- **Planned Value (PV):** Planned Value is the **estimated cost of the work that was planned to be completed** within a specific time according to the project schedule.
- **Earned Value (EV):** Earned Value represents the **value of the work that has actually been completed** at a particular point in the project.
- **Actual Cost (AC):** Actual Cost is the **total amount of money spent** on the work completed in the project.
- **Cost Variance (CV):** Cost Variance shows the **difference between earned value and actual cost** to determine whether the project is over or under budget.
- **Schedule Variance (SV):** Schedule Variance shows the **difference between earned value and planned value** to determine whether the project is ahead or behind schedule.
- **Cost Performance Index (CPI):** CPI measures the **cost efficiency of the project** by comparing earned value with actual cost.
- **Schedule Performance Index (SPI):** SPI measures the **schedule efficiency** of the project by comparing earned value with planned value.

8. Describe the Life Cycle of Project Audit and responsibilities of a project auditor.

A **project audit** is an independent review of a project's processes, controls, and performance at any stage of its life cycle. It helps evaluate compliance, performance, and risks, and provides recommendations for improvement.

Life Cycle of a Project Audit

1. **Initiation / Planning :** Define audit objectives, scope, and criteria such as cost, schedule, quality, risk, and compliance. Select the audit team and prepare an audit plan and checklist.
2. **Data Collection and Baseline Review :** Collect project documents such as project charter, plans, schedules, budgets, risk register, contracts, and progress reports. Establish a baseline to compare planned and actual performance.
3. **Execution: Testing, Analysis, and Evaluation :** Conduct interviews with project teams and stakeholders. Review financial records and compare actual results with planned targets for cost, time, scope, and quality.
4. **Reporting :** Document findings including strengths, weaknesses, risks, and non-conformities. Prepare an audit report with conclusions and recommendations.
5. **Follow-up and Closure :** Monitor whether recommended actions are implemented and verify that corrective measures are effective before closing the audit.

Responsibilities of a Project Auditor

- **Assessing Compliance :** Ensure the project follows laws, regulations, contracts, and organizational policies.
- **Evaluating Project Performance :** Review cost, schedule, scope, and quality against approved project plans.
- **Identifying Risks and Weaknesses :** Detect gaps in controls, documentation, and risk management.
- **Reporting Findings and Recommendations :** Provide clear audit reports with practical improvement suggestions.

- **Follow-up and Accountability** : Track implementation of corrective actions to strengthen project governance.

9. Discuss reasons for project termination and explain the termination process.

Reasons for Project Termination

- **Project Objectives Achieved**: The project is terminated when the **goals and objectives of the project are successfully completed**.
- **Lack of Resources**: A project may be terminated due to **insufficient funds, manpower, or materials** required to continue the project.
- **Poor Project Performance**: If the project is **over budget, delayed, or not meeting quality standards**, management may decide to terminate it.
- **Change in Business Strategy**: Sometimes projects are stopped because of **changes in organizational goals or market conditions**.
- **Technological Changes**: A project may become **obsolete due to new technology or better alternatives**.
- **Legal or Regulatory Issues**: Projects may be terminated if they **violate legal regulations or compliance requirements**.

Project Termination Process

- **Evaluation of Project Status**: Management reviews the **project performance, objectives, and current situation** to decide whether termination is necessary.
- **Decision for Termination**: After evaluation, a **formal decision is made to stop the project**.
- **Documentation and Reporting**: All **project documents, reports, and records** are completed and **archived** for future reference.
- **Release of Resources**: Project resources such as **team members, equipment, and materials** are **reassigned or released**.
- **Final Review and Closure**: A final review is conducted to **analyze project outcomes and capture lessons learned** for future projects.

10. Explain project leadership and challenges in multicultural and virtual teams.

Project Leadership

- Project leadership is the ability of a project manager to **guide, motivate, and influence the project team to achieve project objectives**.
- A project leader is responsible for **planning, decision-making, communication, and problem solving** during the project.
- Effective project leadership helps in **building teamwork, resolving conflicts, and improving team performance**.
- It also involves **encouraging collaboration, innovation, and accountability among team members**.

Challenges in Multicultural and Virtual Teams

- **Cultural Differences**: Team members from different cultures may have **different values, communication styles, and work practices**, which can create misunderstandings.
- **Communication Barriers**: Differences in language and communication styles can make **information sharing and understanding difficult**.
- **Time Zone Differences**: Virtual teams working in different time zones may face **delays in**

communication and coordination.

- **Lack of Face-to-Face Interaction:** Virtual teams may experience **less personal interaction**, which can affect trust and team bonding.
 - **Technology Issues:** Virtual teams depend on **internet and communication tools**, and technical problems can affect project work.
 - **Team Coordination Problems:** Managing and coordinating tasks among **geographically distributed team members** can be challenging.
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(Ab iske baad hamara Internal Assessment nahi hoga, Atharva College me. 🎓)

(Good Bye 🙋)